ANANYA DUBEY

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SUMMARY

Early-career AI/ML engineer building and evaluating LLM, RAG and event-driven backend systems in Python. Built a transformer-embedding RAG assistant with confidence scoring and human review, a Kafka-based security analytics platform with an Ollama LLM assistant, a prompt-injection detection benchmark, and interpretability research on sparse-autoencoder feature stability.

TECHNICAL SKILLS

Backend & Integration: Python, FastAPI, Flask, REST APIs, PostgreSQL, SQL, Redis, Apache Kafka, JWT/RBAC, React, Docker, Git, GitHub Actions

LLM & AI Systems: RAG (FAISS), transformer embedding models, Ollama (local LLMs), prompt engineering, agentic AI workflows, LLM and model evaluation; Claude, Gemini, NVIDIA NIM, Cohere embeddings

ML, Testing & Observability: scikit-learn, PyTorch, TransformerLens, XGBoost, pytest, Prometheus, Grafana, structured logging

EXPERIENCE

AI/Data Intern, ValueMomentum (BFSI domain) Jul 2025 – Mar 2026

TPRA AI Assistant (Flask, FAISS, transformer embeddings)

- Built an end-to-end assistant that drafts Third-Party Risk Assessment questionnaire answers with source citations, using a PDF RAG pipeline (OCR, chunking, MiniLM embeddings, FAISS) and a five-model transformer embedding ensemble for retrieval and scoring.
- Integrated four enterprise data-source connectors (SharePoint, SQL, REST API, file share) through a configurable registry, served via a Flask REST API with web dashboards.
- Implemented confidence scoring with selective human review (auto-approve, spot-check, full-review and reject tiers, with stricter routing for sensitive questions), an audit trail of sources and reviewer decisions, and a feedback loop; 100+ pytest unit tests.
- Projected ~75% reduction in effort per questionnaire (from 40–60 to 10–15 SME hours); demonstrated on 560 mock records, pending enterprise credentials.

Cybersecurity Log Analysis Platform (FastAPI, Kafka, PostgreSQL, Redis)

- Built an event-driven security log analysis platform from FastAPI microservices connected by Apache Kafka, with PostgreSQL, Redis caching and Docker Compose deployment; validated on a 137,611-event Sophos EDR export.
- Implemented a four-layer detection engine (IOC hash matching benchmarked at 1.66M lookups/sec at 100K IOCs, scikit-learn IsolationForest/DBSCAN, behavioural analytics, MITRE ATT&CK mapping) and incident correlation with Union-Find deduplication.
- Added JWT authentication with role-based access control, rate limiting, dead-letter queues, structured JSON logging with correlation IDs and Prometheus metrics.
- Built an Ollama-based LLM assistant for incident summaries and analyst Q&A, and an analyst feedback loop that adjusts detection confidence.

Technical Associate, Nuvanaah Jan 2026 – May 2026

- Developed Nuvanaah's full-stack web platform as part of the company's technical team.

PROJECTS

Life Agent – personal productivity automation system Python, Tkinter, Notion API, GitHub Actions

- Built a multi-provider LLM agent (Claude, Gemini, NVIDIA NIM) integrated with Notion and a local-to-cloud event sync pipeline on GitHub Actions; developed under a structured agent-code-review process.

HairCraft AI – prototype built for K-Hair India (not deployed) Python

- Designed a recommendation system that matches users to hair solutions using confidence scoring and safety constraints, escalating sensitive or medical cases to human experts.

Prompt Injection Detection Benchmark Cohere embed-v4, scikit-learn

- Built a 110-prompt benchmark across 7 attack categories with canary-string detection; trained a logistic-regression detector on Cohere embed-v4 embeddings, evaluated with 5-fold cross-validation.

Fragile Features – interpretability research PyTorch, TransformerLens, SAEBench

- Investigated the stability of sparse-autoencoder features across training seeds using Hungarian matching and SAEBench.

EDUCATION

B.E., Electronics and Computer Science (Honours in AI/ML), VESIT, University of Mumbai 2026

CGPA: 8.53 / 10